

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018- 2019 (Odd Semester)

Innovative Practices Description

UNIT II COUPLING MECHANISM

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC6011 Electromagnetic Interference and Compatibility

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic: Automotive transients

Name of the Innovative Practice: Hardware Demonstration

Description of Hardware Demonstration:

A method demonstration is a teaching method used to communicate an idea with the aid of visuals such as flip charts, posters, power point, etc. A demonstration is the process of teaching someone how to make or do something in a step-by-step process.

The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the generation interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

Outcomes

- The students will be able to analyze the fusing networks and isolators after demonstrating the Fuse networks and isolators.
- The students will be able to identify the advantages of using Miniature Circuit Breaker (MCB) instead of using fusing networks in domestic and as well as industrial environments.

Actually, students are not aware of automotive transients. In real time, they have used automotive transients in their homes unknowingly. For that, I have chosen hardware demonstration method to create the awareness of the automotive transients with the real time applications to the students.

While doing the demonstration, students identified the advantages of using Miniature Circuit Breaker (MCB) instead of using fusing networks in domestic and as well as industrial environments. The assessment of effectiveness of the activity was felt when asked Viva questions after demonstration.

The students are not aware of automotive transient protection devices. During the theory class, some of the students are not interested to listen the class. So, I have shown MCB, Fuse and isolators as automotive transient devices. After demonstrated about Isolators and MCB's, students realized about the necessities of automotive transient protection devices.

Different types of Miniature Circuit Breaker (MCB):



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	3	2	1					1	1		

CO2: The student will be able to measure emission immunity level from different systems to couple with the prescribed EMC standards

References:

1. Clayton Paul, "Introduction to Electromagnetic Compatibility", Wiley Interscience, 2006.
2. V Prasad Kodali, "Engineering Electromagnetic Compatibility", IEEE Press, Newyork, 2001.
3. <https://www.youtube.com/watch?v=9Xgn40eGcqY>

Signature of the Faculty Member

HOD/ECE